

Answer ALL questions, there is only one correct option, No Marks for partial Answers, no negative marking. Assume all header file included

Admission Number and Section: _____

Name: _____

1. Consider the following variable declarations and definitions in C:

- i) `int var_9 = 1;`
- ii) `int 9_var = 2;`
- iii) `int _ = 3;`

Which of the following statement(s) is/are true?

- A. Both i) and ii) are valid
- B. Both i) and iii) are valid**
- C. Only i) is valid
- D. All are valid

2. Consider the following code snippet:

```
int main(){
    int i = 3;
    switch (i){
        case 1:
            printf("NCSV101");
            break;
        case 1+2:
            printf("Quiz 1");
            break;
        default:
            printf("C
                programming");
    }
    return 0;}
```

What is the output of this code snippet?

- A. NCSV101
- B. C programming
- C. Quiz 1**
- D. Compile-time error

3. What is the output of the following code snippet?

```
int main( ) {
    unsigned int a = 0b1001;
    unsigned int b = 0b1100;
    printf("%d", a^b);
    return 0;
}
```

- A. 5**
- B. 0b0101

C. 0101

D. Compilation error

4. Which loop is guaranteed to run for at least once?

- A. for loop
- B. do-while loop**
- C. while loop
- D. none of the above

5. The correct alternative of the following if-else statement is:

```
if(a > b){
    max = a;
} else {
    max = b;}
```

A. `max = a ? a>b : max = b`

B. `max = b ? a>b : max = a`

C. `max = a ? max = b : a>b`

D. `a>b ? max = a : max = b`

6. What does a parallelogram represent in a flowchart?

- A. Process step
- B. Start/End
- C. Input/Output**
- D. Decision

7. What is the result of `5 + 3 * 2 / 4` in C?

- A. 4
- B. 6**
- C. 6.5
- D. 5

8. How many times will the loop execute?

```
for(int i=0; i<5; i+=2) { }
```

- A. 2
- B. 4
- C. 3**
- D. 5

9. What does `continue` do in a loop?

A. Skips the current iteration and proceeds to the next.

B. Exits the loop entirely.

C. Restarts the loop from the beginning.

D. Pauses the loop.

10. What is the output of the following code?

```
int main(){
int a = 5, b = 0;
do {
    b += a;
    a--;
} while (a-- > 0);
printf("\%d", b);
return 0;}
```

A. 15 B. 10 **C. 9** D. 5

11. The output of the following C program is:

```
int main( )
{
int x = 2, y = 5;
(x & y)?printf("A "):printf("B ");
(x && y)?printf("C "):printf("D ");
return 0;
}
```

A. A C B. A D **C. B C** D. B D

12. The output of the following C program is:

```
int main ( )
{
int a = 6;
if ((++a < 7 && (a++ /6))||(++a <= 9))
    printf ( "a=%d " , a );
printf ( "a=%d" , a );
return 0;
}
```

Answer: a=8 a=8

13. The output of the following C program is:

Not correct option given, everyone get marks

```
int main( )
{
int i = 1;
switch( i )
{
    printf ( " My City is ");
    case 1 : printf ("Kolkata"); break;
    case 2 : printf ("Mumbai"); break;
    default : printf ("None of these");
}
return 0;
}
```

A. My City is Kolkata

B. Compilation Error

C. My City is Mumbai

D. None of these

14. The output of the following C program is:

```
int main( )
{
int a = 1;
while(a<=1)
    if(a%2)
        printf("%d ", a++);
printf("%d", a+10);
return 0;
}
```

Answer: 1 12

15. Evaluate the expression below (in standard C) if x = 6 and y = 3 initially:

```
// x = 6 (binary 0110)
//y = 3 (binary 0011)
int result=(x&y) << 1 | (x^y) >> 1;
```

Therefore, the final result is:

A. 5 **B. 6** C. 7 D. 4

16. What is printed by the following code?

```
int main() {
    int n = 1;
    switch (n) {
        case 0: printf("Zero ");
        case 1: printf("One ");
        case 2: printf("Two ");
                break;
        default: printf("Default ");
    }
    return 0;
}
```

A. One Two

B. One Two Default

C. Zero One Two

D. One Two Default followed by a runtime error

17. What will be the output of the following code?

```
int main() {
    int a = 10, b;
    b = (a = a + 5, a * 2);
    printf("%d\n", b);
    return 0;
}
```

A. 10 B. 15 C. 20 **D. 30**

18. What happens in the following code? **Not correct option given, everyone get marks**

```
int main() {
    int x = 1, y = 5;
    while ((x++ < 5) && (y-- > 0)) {
        // empty body
    }
    printf("x = %d, y = %d\n", x, y);
    return 0;
}
```

How many iterations occur, and what are the final values of x and y?

A. The loop runs forever B. 5 iterations; final x = 6, y = 0
C. 4 iterations; final x = 5, y = 1 D. 5 iterations; final x = 5, y = 0

19. How many times does printf("Hello\n"); execute?

```
int main() {
    int i, j;
    for (i = 0; i < 3; i++) {
        for (j = 0; j < 3; j++) {
            if (i == 1) break;
            if (j == 1) continue;
            printf("Hello\n");
        }
    }
    return 0;
}
```

A. 3 **B. 4** C. 5 D. 6

20. The output of the following C program is:

```
int main( )
{
    int a = 0, b = -1, c = 1, res;
    res = !(a) || ++b && ++c ;
    printf("%d %d %d %d", a, b, c, res) ;
    return 0 ;
}
```

A. 0 0 2 0 **B. 0 0 1 0** C. 0 0 1 1 D. Compilation Error

Rough Work